

Geometry & Coordinate Geometry

ANSWER KEY



Distance and midpoint

- 1 Find the distance between (1, 2) and (4, 6).

$$d = \sqrt{(4 - 1)^2 + (6 - 2)^2} = \sqrt{9 + 16} = \sqrt{25} = 5.$$

- 2 Find the midpoint of (2, -3) and (8, 5).

$$\text{Midpoint} = \left(\frac{2 + 8}{2}, \frac{-3 + 5}{2} \right) = (5, 1).$$

- 3 Find the distance between (-2, 1) and (3, -11).

$$d = \sqrt{(3 - (-2))^2 + (-11 - 1)^2} = \sqrt{25 + 144} = \sqrt{169} = 13.$$

- 4 Find the midpoint of (-4, 7) and (6, -3).

$$\text{Midpoint} = \left(\frac{-4 + 6}{2}, \frac{7 - 3}{2} \right) = (1, 2).$$

Gradient (slope)

- 5 Find the gradient of the line through (1, 2) and (5, 10).

$$m = \frac{10 - 2}{5 - 1} = \frac{8}{4} = 2.$$

- 6 Find the gradient of the line through (-3, 4) and (2, -6).

$$m = \frac{-6 - 4}{2 - (-3)} = \frac{-10}{5} = -2.$$

- 7 Find the gradient of the line through (0, 0) and (6, 9).

$$m = \frac{9 - 0}{6 - 0} = \frac{9}{6} = \frac{3}{2}.$$

- 8 State whether the line through (1, 1) and (4, 1) is horizontal, vertical, or neither.

Both points have $y = 1$, so the line is horizontal (gradient = 0).

Equations of lines

- 9 Find the equation of the line with gradient 2 passing through (3, 5), in the form $y = mx + c$.

$$y - 5 = 2(x - 3) \Rightarrow y = 2x - 1.$$

- 10 Find the equation of the line through (0, 4) and (2, 10).

$$m = \frac{10 - 4}{2 - 0} = 3. \text{ Using the y-intercept 4: } y = 3x + 4.$$

- 11 Find the x- and y-intercepts of the line $2x + 3y = 12$.

$$\text{x-intercept (y = 0): } x = 6. \text{ y-intercept (x = 0): } y = 4.$$

- 12 Write the equation of the line parallel to $y = 4x - 1$ passing through (2, 9).

$$\text{Same gradient 4: } y - 9 = 4(x - 2) \Rightarrow y = 4x + 1.$$

Parallel and perpendicular lines

- 13 State the gradient of a line perpendicular to a line with gradient 5.

$$m = -\frac{1}{5}.$$

- 14 Determine whether the lines $y = 2x + 3$ and $y = 2x - 7$ are parallel, perpendicular, or neither.

Parallel — both have gradient 2.

- 15 Find the equation of the line perpendicular to $y = -\frac{1}{3}x + 2$, passing through (3, 4).

$$\text{The perpendicular gradient is 3. } y - 4 = 3(x - 3) \Rightarrow y = 3x - 5.$$

The equation of a circle

- 16 Write the equation of a circle centered at the origin with radius 6.

$$x^2 + y^2 = 36.$$

- 17 Write the equation of a circle centered at (2, -3) with radius 5.

$$(x - 2)^2 + (y + 3)^2 = 25.$$

- 18 Find the radius and center of the circle $(x - 1)^2 + (y + 4)^2 = 49$.

Center: (1, -4). Radius: 7.

- 19 Find the equation of the circle centered at the origin that passes through (3, 4).

$$\text{Radius} = \sqrt{3^2 + 4^2} = 5. \text{ Equation: } x^2 + y^2 = 25.$$

Circles: chords and arcs

- 20 Points A and B lie on a circle of radius 5 centered at O , with central angle $\angle AOB = 90^\circ$. Find the length of chord AB , using $AB = 2r \sin\left(\frac{\theta}{2}\right)$.

$$AB = 2(5)\sin(45^\circ) = 10(0.7071) \approx 7.07.$$

- 21 A chord of a circle with radius 10 cm is 12 cm long. Find the distance from the center to the chord.

$$\text{Half the chord is 6 cm. Distance} = \sqrt{10^2 - 6^2} = \sqrt{100 - 36} = \sqrt{64} = 8 \text{ cm.}$$

- 22 Find the circumference of a circle with radius 9 cm, in terms of π .

$$C = 2\pi r = 18\pi \text{ cm.}$$

Polygon angle sums

- 23 Find the sum of the interior angles of a hexagon (6 sides).

$$(n - 2)180^\circ = (6 - 2)180^\circ = 720^\circ.$$

- 24 Find the measure of each interior angle of a regular pentagon.

$$\text{Sum} = (5 - 2)180^\circ = 540^\circ. \text{ Each angle} = \frac{540^\circ}{5} = 108^\circ.$$

- 25 A polygon has an interior angle sum of 1440° . Find the number of sides.

$$(n - 2)180^\circ = 1440^\circ \Rightarrow n - 2 = 8 \Rightarrow n = 10.$$

Perimeter and area

- 26 A rectangle has length 12 and width 7. Find its perimeter and area.

$$\text{Perimeter} = 2(12 + 7) = 38. \text{ Area} = 12 \times 7 = 84.$$

- 27 A right triangle has legs 6 and 8. Find its area and hypotenuse.

$$\text{Area} = \frac{1}{2}(6)(8) = 24. \text{ Hypotenuse} = \sqrt{36 + 64} = \sqrt{100} = 10.$$

- 28 A regular hexagon has side length 4, with area $A = \frac{3\sqrt{3}}{2}s^2$. Find its area in exact form.

$$A = \frac{3\sqrt{3}}{2}(16) = 24\sqrt{3}.$$

Coordinate geometry: identifying shapes

- 29 Show that the triangle with vertices $A(0, 0)$, $B(4, 0)$, $C(4, 3)$ is a right triangle.

AB is horizontal and BC is vertical, so they are perpendicular; the right angle is at B .

- 30** Determine whether the points $A(-2, 1)$, $B(2, 4)$, $C(5, 0)$ form a right triangle, using the squared side lengths.

$AB^2 = 4^2 + 3^2 = 25$. $BC^2 = 3^2 + 4^2 = 25$. $AC^2 = 7^2 + 1^2 = 50$. Since $AB^2 + BC^2 = 50 = AC^2$, the triangle has a right angle at B .

- 31** Determine whether the triangle with vertices $A(1, 2)$, $B(5, 2)$, $C(3, 6)$ is isosceles.

$AC = \sqrt{2^2 + 4^2} = \sqrt{20}$ and $BC = \sqrt{2^2 + 4^2} = \sqrt{20}$, so $AC = BC$ and the triangle is isosceles.

The perpendicular bisector of a segment

- 32** Find the equation of the perpendicular bisector of the segment joining $(2, 3)$ and $(8, 7)$.

Midpoint $= (5, 5)$. Segment gradient $= \frac{7-3}{8-2} = \frac{2}{3}$, so the perpendicular gradient is $-\frac{3}{2}$. Equation:
 $y - 5 = -\frac{3}{2}(x - 5) \Rightarrow y = -\frac{3}{2}x + \frac{25}{2}$.

- 33** Find the equation of the perpendicular bisector of the segment joining $(0, 0)$ and $(6, 8)$.

Midpoint $= (3, 4)$. Segment gradient $= \frac{8}{6} = \frac{4}{3}$, so the perpendicular gradient is $-\frac{3}{4}$. Equation:
 $y - 4 = -\frac{3}{4}(x - 3) \Rightarrow y = -\frac{3}{4}x + \frac{25}{4}$.

- 34** A line segment has endpoints $(1, 5)$ and $(7, 5)$. Find the equation of its perpendicular bisector.

Midpoint $= (4, 5)$. The segment is horizontal (gradient 0), so the perpendicular bisector is the vertical line $x = 4$.

Word problems: geometry on a coordinate grid

- 35** This question has two parts. A city park is designed on a coordinate grid (in meters), with two gates at $A(0, 0)$ and $B(80, 60)$.

a Find the straight-line distance between the two gates.

$$AB = \sqrt{80^2 + 60^2} = \sqrt{6400 + 3600} = \sqrt{10000} = 100 \text{ meters.}$$

b A fountain is placed at the midpoint between the two gates. Find its coordinates. Midpoint
 $= \left(\frac{0+80}{2}, \frac{0+60}{2} \right) = (40, 30)$.

- 36** This question has two parts. A circular irrigation sprinkler is centered at $(5, 5)$ on a farm's coordinate grid (in meters) and has a radius of 12 meters.

a Write the equation of the circle covered by the sprinkler. $(x - 5)^2 + (y - 5)^2 = 144$.

b Determine whether a tree at $(14, 9)$ lies inside, on, or outside the watered region. Distance from center $= \sqrt{(14 - 5)^2 + (9 - 5)^2} = \sqrt{81 + 16} = \sqrt{97} \approx 9.85$ meters, which is less than the radius 12, so the tree lies inside the watered region.

MCQ — Distance and midpoint

37 Find the distance between (2, 3) and (6, 6).

a. 5 ✓

b. 25

c. $\sqrt{7}$

d. 7

$$\sqrt{(6-2)^2 + (6-3)^2} = \sqrt{16+9} = 5.$$

38 Find the midpoint of (3, -4) and (9, 6).

a. (6, 1) ✓

b. (6, 2)

c. (12, 2)

d. (3, 10)

$$\left(\frac{3+9}{2}, \frac{-4+6}{2}\right) = (6, 1).$$

39 Find the distance between (-3, 2) and (5, -13).

a. 17 ✓

b. 15

c. 19

d. 289

$$\sqrt{8^2 + 15^2} = \sqrt{64 + 225} = \sqrt{289} = 17.$$

40 Find the midpoint of $(-5, 8)$ and $(7, -2)$.

a. $(6, 1)$

b. $(1, 6)$

c. $(2, 6)$

d. $(1, 3)$ ✓

$$\left(\frac{-5 + 7}{2}, \frac{8 - 2}{2} \right) = (1, 3).$$

41 Find the distance between the two points shown.

a. $\sqrt{5}$

b. 4

c. 3

d. 5 ✓

$$\sqrt{(5 - 1)^2 + (5 - 2)^2} = \sqrt{16 + 9} = 5.$$

42 Find the midpoint of the segment joining the two points shown.

a. $(4, 3)$

b. $(2, 6)$

c. $(2, 3)$ ✓

d. $(8, 6)$

$$\left(\frac{-2 + 6}{2}, \frac{3 + 3}{2} \right) = (2, 3).$$

MCQ — Gradient (slope)

43 Find the gradient of the line through (2, 3) and (6, 11).

a. 8

b. 0.5

c. 4

d. 2 ✓

$$m = \frac{11 - 3}{6 - 2} = 2.$$

44 Find the gradient of the line through (−2, 5) and (3, −10).

a. $-\frac{1}{3}$

b. −15

c. −3 ✓

d. 3

$$m = \frac{-10 - 5}{3 - (-2)} = \frac{-15}{5} = -3.$$

45 Find the gradient of the line through (0, 0) and (8, 12).

a. 1.5 ✓

b. $\frac{8}{12}$

c. 96

d. $\frac{2}{3}$

$$m = \frac{12}{8} = 1.5.$$

46 State whether the line through (2, 3) and (2, 9) is horizontal, vertical, or neither.

a. Cannot be determined

b. Neither

c. Horizontal

d. Vertical ✓

Both points share $x = 2$: the line is vertical.

MCQ — Equations of lines

47 Find the equation of the line with gradient 3 passing through (2, 4), in the form $y = mx + c$.

a. $y = 3x + 4$

b. $y = 3x - 2$ ✓

c. $y = 3x + 2$

d. $y = 2x - 3$

$$4 = 3(2) + c \Rightarrow c = -2.$$

48 Find the equation of the line through (0, 5) and (3, 14).

a. $y = 5x + 3$

b. $y = 9x + 5$

c. $y = 3x + 5$ ✓

d. $y = 3x - 5$

$$m = \frac{14 - 5}{3} = 3; c = 5 \text{ (the } y\text{-intercept).}$$

49 Find the x- and y-intercepts of the line $3x + 4y = 24$.

a. $x = 8, y = 6$ ✓

b. $x = 6, y = 8$

c. $x = 24, y = 24$

d. $x = 4, y = 3$

$$\text{Set } y = 0: x = 8. \text{ Set } x = 0: y = 6.$$

50 Write the equation of the line parallel to $y = 5x - 2$ passing through $(1, 8)$.

a. $y = 5x + 3$ ✓

b. $y = -\frac{1}{5}x + 3$

c. $y = 5x - 3$

d. $y = 5x + 8$

$$8 = 5(1) + c \Rightarrow c = 3.$$

51 Find the equation of the line through $(1, 3)$ and $(4, 12)$.

a. $y = 3x + 3$

b. $y = x + 3$

c. $y = 3x - 3$

d. $y = 3x$ ✓

$$m = \frac{12 - 3}{4 - 1} = 3; 3 = 3(1) + c \Rightarrow c = 0.$$

52 The graph shows a line. Find its equation in the form $y = mx + c$ by reading the graph.

a. $y = \frac{1}{2}x + 1$

b. $y = 2x + 1$ ✓

c. $y = x + 1$

d. $y = 2x - 1$

The line crosses the y -axis at 1 and rises 2 for every 1 across: $y = 2x + 1$.

MCQ — Parallel and perpendicular lines

53 State the gradient of a line perpendicular to a line with gradient 4.

a. $-\frac{1}{4}$ ✓

b. 4

c. $\frac{1}{4}$

d. -4

Perpendicular gradients multiply to -1 : $-\frac{1}{4}$.

54 Determine whether the lines $y = 3x + 2$ and $y = 3x - 5$ are parallel, perpendicular, or neither.

a. The same line

b. Neither

c. Perpendicular

d. **Parallel** ✓

Both have gradient 3: parallel.

55 Find the equation of the line perpendicular to $y = -\frac{1}{4}x + 3$, passing through (4, 5).

a. $y = 4x - 5$

b. $y = 4x + 11$

c. $y = -\frac{1}{4}x + 6$

d. $y = 4x - 11$ ✓

Perpendicular gradient = 4: $5 = 4(4) + c \Rightarrow c = -11$.

56 Determine whether the lines $y = 2x + 1$ and $y = -0.5x + 4$ are parallel, perpendicular, or neither.

a. The same line

b. Parallel

c. Neither

d. **Perpendicular** ✓

Gradients 2 and -0.5 multiply to -1 : perpendicular.

MCQ — The equation of a circle

57 Write the equation of a circle centered at the origin with radius 8.

a. $x^2 + y^2 = 64$ ✓

b. $x^2 + y^2 = 16$

c. $x^2 + y^2 = 8$

d. $(x - 8)^2 + (y - 8)^2 = 64$

$r^2 = 64$.

58 Write the equation of a circle centered at $(3, -2)$ with radius 7.

a. $(x - 3)^2 - (y + 2)^2 = 49$

b. $(x - 3)^2 + (y + 2)^2 = 49$ ✓

c. $(x - 3)^2 + (y + 2)^2 = 7$

d. $(x + 3)^2 + (y - 2)^2 = 49$

$(x - 3)^2 + (y - (-2))^2 = 7^2$.

59 Find the radius and center of the circle $(x + 2)^2 + (y - 5)^2 = 81$.

a. Center $(2, -5)$, radius 9

b. Center $(-2, 5)$, radius 9 ✓

c. Center $(-2, 5)$, radius 81

d. Center $(2, 5)$, radius 9

$(x - (-2))^2 + (y - 5)^2 = 9^2$.

60 Find the equation of the circle centered at the origin that passes through (6, 8).

a. $x^2 + y^2 = 100$ ✓

b. $x^2 + y^2 = 10$

c. $x^2 + y^2 = 196$

d. $x^2 + y^2 = 48$

$$r = \sqrt{6^2 + 8^2} = 10; r^2 = 100.$$

61 Write the equation of the circle shown, centered at the origin.

a. $x^2 + y^2 = 10$

b. $x^2 + y^2 = 5$

c. $x^2 + y^2 = 20$

d. $x^2 + y^2 = 25$ ✓

$$r = 5; r^2 = 25.$$

62 The circle shown is centered at (1, 2) with the radius marked. Write its equation.

a. $(x - 1)^2 + (y - 2)^2 = 8$

b. $(x + 1)^2 + (y + 2)^2 = 16$

c. $(x - 1)^2 + (y - 2)^2 = 16$ ✓

d. $(x - 1)^2 + (y - 2)^2 = 4$

$$(x - 1)^2 + (y - 2)^2 = 4^2.$$

MCQ — Circles: chords and arcs

- 63 Points A and B lie on a circle of radius 8 centered at O , with central angle $\angle AOB = 60^\circ$. Find the arc length AB , in terms of π .

a. $\frac{16\pi}{3}$

b. $\frac{4\pi}{3}$

c. 16π

d. $\frac{8\pi}{3}$ ✓

$$\text{Arc} = \frac{60}{360} \times 2\pi(8) = \frac{8\pi}{3}.$$

- 64 A chord of a circle with radius 13 cm is 24 cm long. Find the distance from the center to the chord.

a. 7

b. 1

c. 12

d. 5 ✓

$$\text{Half-chord} = 12; \text{distance} = \sqrt{13^2 - 12^2} = \sqrt{25} = 5 \text{ cm.}$$

- 65 Find the circumference of a circle with radius 11 cm, in terms of π .

a. 121π

b. 44π

c. 22π ✓

d. 11π

$$C = 2\pi(11) = 22\pi \text{ cm.}$$

66 Find the arc length of the 90° sector shown, in terms of π .

a. 12π

b. 3π ✓

c. 1.5π

d. 6π

$$\text{Arc} = \frac{90}{360} \times 2\pi(6) = 3\pi.$$

MCQ — Polygon angle sums

67 Find the sum of the interior angles of an octagon (8 sides).

a. 720°

b. $1,440^\circ$

c. 900°

d. $1,080^\circ$ ✓

$$(8 - 2) \times 180 = 1,080^\circ.$$

68 Find the measure of each interior angle of a regular hexagon.

a. 140°

b. 135°

c. 108°

d. 120° ✓

$$\text{Sum} = 720^\circ; \text{ each } = 720 \div 6 = 120^\circ.$$

69 A polygon has an interior angle sum of $1,620^\circ$. Find the number of sides.

- a. 10
- b. 12
- c. 9
- d. 11 ✓

$$(n - 2) \times 180 = 1,620 \Rightarrow n - 2 = 9 \Rightarrow n = 11.$$

70 Find the sum of the interior angles of a decagon (10 sides).

- a. $1,800^\circ$
- b. $1,260^\circ$
- c. $1,080^\circ$
- d. $1,440^\circ$ ✓

$$(10 - 2) \times 180 = 1,440^\circ.$$

MCQ — Perimeter and area

71 A rectangle has length 15 and width 9, as shown. Find its perimeter and area.

- a. $P = 135, A = 48$
- b. $P = 48, A = 135$ ✓
- c. $P = 24, A = 135$
- d. $P = 48, A = 24$

$$P = 2(15 + 9) = 48; A = 15 \times 9 = 135.$$

72 A right triangle has legs 5 and 12, as shown. Find its area and hypotenuse.

a. $A = 60$, $\text{hyp} = 13$

b. $A = 30$, $\text{hyp} = 17$

c. $A = 30$, $\text{hyp} = 13$ ✓

d. $A = 30$, $\text{hyp} = 12$

$$A = \frac{1}{2}(5)(12) = 30; \text{hyp} = \sqrt{5^2 + 12^2} = 13.$$

73 A regular hexagon has side length 6, with area $A = \frac{3\sqrt{3}}{2}s^2$. Find its area in exact form.

a. $36\sqrt{3}$

b. $108\sqrt{3}$

c. $54\sqrt{3}$ ✓

d. $18\sqrt{3}$

$$A = \frac{3\sqrt{3}}{2}(36) = 54\sqrt{3}.$$

74 A regular pentagon has side length 8, with area $A \approx 1.72s^2$. Find its area to the nearest whole number.

a. 110 ✓

b. 124

c. 96

d. 14

$$A \approx 1.72(64) \approx 110.$$

MCQ — Coordinate geometry: identifying shapes

75 Is the triangle with vertices $A(0, 0)$, $B(6, 0)$, $C(6, 4)$ a right triangle?

- a. No
- b. Cannot be determined
- c. Yes ✓
- d. Only if it is also isosceles

AB is horizontal and BC is vertical, so they meet at a right angle at B : yes.

76 Determine whether the points $A(-3, 2)$, $B(1, 5)$, $C(4, 1)$ form a right triangle, using squared distances.

- a. Only if it is also equilateral
- b. No
- c. Cannot be determined
- d. Yes ✓

$AB^2 = 25$, $BC^2 = 25$, $AC^2 = 50$. Since $AB^2 + BC^2 = AC^2$, it's a right triangle at B : yes.

77 Determine whether the triangle with vertices $A(2, 3)$, $B(6, 3)$, $C(4, 7)$ is isosceles.

- a. No
- b. Only if it is also a right triangle
- c. Yes ✓
- d. Cannot be determined

$AC = \sqrt{4 + 16} = \sqrt{20}$ and $BC = \sqrt{4 + 16} = \sqrt{20}$: two equal sides, so yes.

78 Is the triangle shown a right triangle?

- a. Only if it is also isosceles
- b. Cannot be determined
- c. No
- d. Yes ✓

The two legs meet at a right angle, as shown.

79 Is the triangle shown isosceles, scalene, or equilateral?

- a. Isosceles ✓
- b. Cannot be determined
- c. Equilateral
- d. Scalene

Two of the three sides shown are equal in length: isosceles.

MCQ — The perpendicular bisector of a segment

80 Find the equation of the perpendicular bisector of the segment joining (1, 4) and (7, 10).

- a. $y = -x + 11$ ✓
- b. $y = x - 11$
- c. $y = x + 11$
- d. $y = -x + 7$

Midpoint = (4, 7); segment slope = 1; perpendicular slope = -1: $y - 7 = -(x - 4)$.

81 A line segment has endpoints (2, 0) and (2, 8). Find the equation of its perpendicular bisector.

- a. $x = 4$
- b. $y = 4$ ✓
- c. $y = 2$
- d. $x = 2$

Midpoint = (2, 4); the segment is vertical, so the bisector is horizontal: $y = 4$.

82 A line segment has endpoints (3, 6) and (9, 6). Find the equation of its perpendicular bisector.

- a. $x = 3$
- b. $y = 6$
- c. $y = 9$
- d. $x = 6$ ✓

Midpoint = (6, 6); the segment is horizontal, so the bisector is vertical: $x = 6$.

83 Find the equation of the perpendicular bisector of the segment joining (−2, 1) and (4, 9).

- a. $y = -\frac{3}{4}x + \frac{23}{4}$ ✓
- b. $y = -\frac{3}{4}x + 5$
- c. $y = -\frac{4}{3}x + \frac{23}{4}$
- d. $y = \frac{3}{4}x + \frac{23}{4}$

Midpoint = (1, 5); segment slope = $\frac{4}{3}$; perpendicular slope = $-\frac{3}{4}$.

MCQ — Word problems: geometry on a coordinate grid

84 A city park is designed on a coordinate grid (in meters). Two gates are located at (10, 20) and (50, 60). Find the distance between the gates, to the nearest meter.

- a. 113
- b. 40
- c. 57 ✓
- d. 80

$$\sqrt{40^2 + 40^2} = 40\sqrt{2} \approx 57 \text{ m.}$$

- 85 A circular irrigation sprinkler is centered at $(8, 8)$ on a farm's coordinate grid (in meters) and has a radius of 15 meters. Write the equation of the circle it waters.

a. $(x + 8)^2 + (y + 8)^2 = 225$

b. $(x - 8)^2 + (y - 8)^2 = 15$

c. $(x - 8)^2 + (y - 8)^2 = 225$ ✓

d. $(x - 8)^2 + (y - 8)^2 = 30$

$$(x - 8)^2 + (y - 8)^2 = 15^2.$$

- 86 A surveyor marks two points on a coordinate map (in km): a base camp at $(3, 4)$ and a summit at $(15, 4)$. Find the midpoint, to be used as a resupply station.

a. $(6, 4)$

b. $(9, 8)$

c. $(9, 4)$ ✓

d. $(18, 8)$

$$\left(\frac{3 + 15}{2}, \frac{4 + 4}{2} \right) = (9, 4).$$

Multiple choice — easy

- 87 Find the distance between $(3, 2)$ and $(-2, 14)$.

a. 15

b. 13 ✓

c. 17

d. 11

$$d = \sqrt{(-2 - 3)^2 + (14 - 2)^2} = \sqrt{25 + 144} = \sqrt{169} = 13.$$

88 Find the distance between $(-7, -8)$ and $(2, 4)$.

a. 15 ✓

b. 17

c. 21

d. 13

$$d = \sqrt{(2 - (-7))^2 + (4 - (-8))^2} = \sqrt{81 + 144} = \sqrt{225} = 15.$$

89 Find the distance between $(-5, 7)$ and $(-17, 23)$.

a. 22

b. 28

c. 20 ✓

d. 18

$$d = \sqrt{(-17 - (-5))^2 + (23 - 7)^2} = \sqrt{144 + 256} = \sqrt{400} = 20.$$

90 Find the distance between $(7, 4)$ and $(4, 0)$.

a. 8

b. 3

c. 7

d. 5 ✓

$$d = \sqrt{(4 - 7)^2 + (0 - 4)^2} = \sqrt{9 + 16} = \sqrt{25} = 5.$$

91 Find the distance between $(4, 0)$ and $(7, -4)$.

a. 5 ✓

b. 3

c. 7

d. 8

$$d = \sqrt{(7 - 4)^2 + (-4 - 0)^2} = \sqrt{9 + 16} = \sqrt{25} = 5.$$

92 Find the midpoint of (7, 6) and (−9, 2).

- a. (0, 4)
- b. (0, 5)
- c. (−1, 5)
- d. (−1, 4) ✓

$$\text{Midpoint} = \left(\frac{7 + (-9)}{2}, \frac{6 + 2}{2} \right) = (-1, 4).$$

93 Find the midpoint of (8, 3) and (0, −1).

- a. (4, 2)
- b. (4, 1) ✓
- c. (5, 2)
- d. (5, 1)

$$\text{Midpoint} = \left(\frac{8 + 0}{2}, \frac{3 + (-1)}{2} \right) = (4, 1).$$

94 Find the midpoint of (−1, 0) and (−1, 2).

- a. (0, 2)
- b. (−1, 1) ✓
- c. (0, 1)
- d. (−1, 2)

$$\text{Midpoint} = \left(\frac{-1 + (-1)}{2}, \frac{0 + 2}{2} \right) = (-1, 1).$$

95 Find the midpoint of $(-6, 7)$ and $(-6, 1)$.

- a. $(-5, 5)$
- b. $(-5, 4)$
- c. $(-6, 5)$
- d. $(-6, 4)$ ✓

$$\text{Midpoint} = \left(\frac{-6 - 6}{2}, \frac{7 + 1}{2} \right) = (-6, 4).$$

96 Find the midpoint of $(-5, 2)$ and $(9, -4)$.

- a. $(2, -1)$ ✓
- b. $(3, 0)$
- c. $(3, -1)$
- d. $(2, 0)$

$$\text{Midpoint} = \left(\frac{-5 + 9}{2}, \frac{2 - 4}{2} \right) = (2, -1).$$

97 Find the gradient of the line through $(-3, 2)$ and $(-6, 3)$.

- a. -3
- b. $-\frac{1}{3}$ ✓
- c. $\frac{1}{3}$
- d. -1

$$m = \frac{3 - 2}{-6 - (-3)} = \frac{1}{-3} = -\frac{1}{3}.$$

98 Find the gradient of the line through $(-5, 2)$ and $(-3, 0)$.

- a. 0
- b. $-\frac{1}{2}$
- c. -1 ✓
- d. 1

$$m = \frac{0 - 2}{-3 - (-5)} = \frac{-2}{2} = -1.$$

99 Find the gradient of the line through $(-5, -5)$ and $(6, 17)$.

a. $\frac{1}{2}$

b. $\frac{24}{11}$

c. 2 ✓

d. -2

$$m = \frac{17 - (-5)}{6 - (-5)} = \frac{22}{11} = 2.$$

100 Find the gradient of the line through $(6, 1)$ and $(5, 2)$.

a. -3

b. -2

c. -1 ✓

d. 1

$$m = \frac{2 - 1}{5 - 6} = \frac{1}{-1} = -1.$$

101 Find the gradient of the line through $(0, 1)$ and $(5, 21)$.

a. -4

b. $\frac{1}{4}$

c. 4 ✓

d. $\frac{22}{5}$

$$m = \frac{21 - 1}{5 - 0} = \frac{20}{5} = 4.$$

Multiple choice — medium

102 Find the equation of the line with gradient 4 passing through $(-4, 3)$, in the form $y = mx + c$.

a. $y = 4x + 21$

b. $y = 4x + 19$ ✓

c. $y = -4x + 19$

d. $y = 4x - 19$

$$y - 3 = 4(x - (-4)) \Rightarrow y = 4x + 19.$$

103 Find the equation of the line with gradient -4 passing through $(-5, -5)$, in the form $y = mx + c$.

a. $y = -4x + 25$

b. $y = -4x - 23$

c. $y = 4x - 25$

d. $y = -4x - 25$ ✓

$$y - (-5) = -4(x - (-5)) \Rightarrow y = -4x - 25.$$

104 Find the equation of the line with gradient -3 passing through $(6, 2)$, in the form $y = mx + c$.

a. $y = -3x + 20$ ✓

b. $y = -3x + 22$

c. $y = 3x + 20$

d. $y = -3x - 20$

$$y - 2 = -3(x - 6) \Rightarrow y = -3x + 20.$$

105 Find the equation of the line with gradient -3 passing through $(-4, 2)$, in the form $y = mx + c$.

a. $y = -3x - 10$ ✓

b. $y = -3x + 10$

c. $y = 3x - 10$

d. $y = -3x - 8$

$$y - 2 = -3(x - (-4)) \Rightarrow y = -3x - 10.$$

106 Find the equation of the line with gradient 2 passing through $(-2, -3)$, in the form $y = mx + c$.

a. $y = 2x + 3$

b. $y = 2x + 1$ ✓

c. $y = -2x + 1$

d. $y = 2x - 1$

$$y - (-3) = 2(x - (-2)) \Rightarrow y = 2x + 1.$$

107 State the gradient of a line perpendicular to a line with gradient 4.

a. $\frac{1}{4}$

b. $-\frac{1}{4}$ ✓

c. 4

d. $-\frac{1}{6}$

$$m = -\frac{1}{4}.$$

108 State the gradient of a line perpendicular to a line with gradient $\frac{4}{3}$.

a. $-\frac{1}{2}$

b. $\frac{4}{3}$

c. $-\frac{3}{4}$ ✓

d. $\frac{3}{4}$

$$m = -\frac{3}{4}.$$

109 State the gradient of a line perpendicular to a line with gradient $\frac{3}{5}$.

a. $-\frac{5}{3}$ ✓

b. -1

c. $\frac{3}{5}$

d. $\frac{5}{3}$

$$m = -\frac{5}{3}.$$

110 State the gradient of a line perpendicular to a line with gradient $\frac{2}{5}$.

a. $\frac{2}{5}$

b. $\frac{5}{2}$

c. $-\frac{5}{4}$

d. $-\frac{5}{2}$ ✓

$$m = -\frac{5}{2}.$$

111 State the gradient of a line perpendicular to a line with gradient $\frac{4}{7}$.

a. $\frac{7}{4}$

b. $-\frac{7}{6}$

c. $-\frac{7}{4}$ ✓

d. $\frac{4}{7}$

$$m = -\frac{7}{4}.$$

112 Write the equation of a circle centered at $(-1, -2)$ with radius 4.

a. $(x - 1)^2 + (y + 2)^2 = 16$

b. $(x + 1)^2 + (y - 2)^2 = 16$

c. $(x + 1)^2 + (y + 2)^2 = 16$ ✓

d. $(x + 1)^2 + (y + 2)^2 = 25$

$$(x + 1)^2 + (y + 2)^2 = 16.$$

113 Write the equation of a circle centered at $(-6, -2)$ with radius 3.

a. $(x + 6)^2 + (y + 2)^2 = 16$

b. $(x - 6)^2 + (y + 2)^2 = 9$

c. $(x + 6)^2 + (y - 2)^2 = 9$

d. $(x + 6)^2 + (y + 2)^2 = 9$ ✓

$(x + 6)^2 + (y + 2)^2 = 9.$

114 Write the equation of a circle centered at $(-4, 5)$ with radius 3.

a. $(x + 4)^2 + (y - 5)^2 = 9$ ✓

b. $(x - 4)^2 + (y - 5)^2 = 9$

c. $(x + 4)^2 + (y - 5)^2 = 16$

d. $(x + 4)^2 + (y + 5)^2 = 9$

$(x + 4)^2 + (y - 5)^2 = 9.$

115 Write the equation of a circle centered at $(1, 1)$ with radius 7.

a. $(x - 1)^2 + (y + 1)^2 = 49$

b. $(x - 1)^2 + (y - 1)^2 = 64$

c. $(x + 1)^2 + (y - 1)^2 = 49$

d. $(x - 1)^2 + (y - 1)^2 = 49$ ✓

$(x - 1)^2 + (y - 1)^2 = 49.$

116 Write the equation of a circle centered at the origin with radius 3.

a. $x^2 + (y - 1)^2 = 9$

b. $x^2 + y^2 = 16$

c. $x^2 + y^2 = 9$ ✓

d. $(x - 1)^2 + y^2 = 9$

$x^2 + y^2 = 9$.

Multiple choice — hard

117 Find the sum of the interior angles of a nonagon (9 sides).

a. 1440°

b. 1620°

c. 1260° ✓

d. 1080°

$(n - 2)180^\circ = (9 - 2)180^\circ = 1260^\circ$.

118 Find the measure of each interior angle of a regular dodecagon.

a. 30°

b. 155°

c. 150° ✓

d. 145°

Sum = $(12 - 2)180^\circ = 1800^\circ$. Each angle = $\frac{1800^\circ}{12} = 150^\circ$.

119 Find the measure of each interior angle of a regular nonagon.

- a. 145°
- b. 140° ✓
- c. 135°
- d. 40°

$$\text{Sum} = (9 - 2)180^\circ = 1260^\circ. \text{ Each angle} = \frac{1260^\circ}{9} = 140^\circ.$$

120 Find the sum of the interior angles of an octagon (8 sides).

- a. 1440°
- b. 1260°
- c. 1080° ✓
- d. 900°

$$(n - 2)180^\circ = (8 - 2)180^\circ = 1080^\circ.$$

121 A polygon has an interior angle sum of 1080° . Find the number of sides.

- a. 10
- b. 9
- c. 7
- d. 8 ✓

$$(n - 2)180^\circ = 1080^\circ \Rightarrow n - 2 = 6 \Rightarrow n = 8.$$

122 A right triangle has legs 7 and 24. Find its area and hypotenuse.

- a. Area: 84. Hypotenuse: 26
- b. Area: 168. Hypotenuse: 25
- c. Area: 91. Hypotenuse: 25
- d. Area: 84. Hypotenuse: 25 ✓

$$\text{Area} = \frac{1}{2}(7)(24) = 84. \text{ Hypotenuse} = \sqrt{49 + 576} = \sqrt{625} = 25.$$

123 A rectangle has length 14 and width 6. Find its perimeter and area.

- a. Perimeter: 20. Area: 84
- b. Perimeter: 40. Area: 98
- c. Perimeter: 42. Area: 84
- d. Perimeter: 40. Area: 84 ✓

$$\text{Perimeter} = 2(14 + 6) = 40. \text{Area} = 14 \times 6 = 84.$$

124 A right triangle has legs 12 and 16. Find its area and hypotenuse.

- a. Area: 96. Hypotenuse: 21
- b. Area: 96. Hypotenuse: 20 ✓
- c. Area: 108. Hypotenuse: 20
- d. Area: 192. Hypotenuse: 20

$$\text{Area} = \frac{1}{2}(12)(16) = 96. \text{Hypotenuse} = \sqrt{144 + 256} = \sqrt{400} = 20.$$

125 A rectangle has length 7 and width 6. Find its perimeter and area.

- a. Perimeter: 28. Area: 42
- b. Perimeter: 26. Area: 49
- c. Perimeter: 26. Area: 42 ✓
- d. Perimeter: 13. Area: 42

$$\text{Perimeter} = 2(7 + 6) = 26. \text{Area} = 7 \times 6 = 42.$$

126 A right triangle has legs 5 and 12. Find its area and hypotenuse.

a. Area: 35. Hypotenuse: 13

b. Area: 60. Hypotenuse: 13

c. Area: 30. Hypotenuse: 13 ✓

d. Area: 30. Hypotenuse: 14

$$\text{Area} = \frac{1}{2}(5)(12) = 30. \text{ Hypotenuse} = \sqrt{25 + 144} = \sqrt{169} = 13.$$

127 Find the equation of the perpendicular bisector of the segment joining $(-1, 4)$ and $(5, 12)$.

a. $y - 2 = -\frac{3}{4}(x - 8)$

b. $y - 8 = \frac{3}{4}(x - 2)$

c. $y - 8 = \frac{4}{3}(x - 2)$

d. $y - 8 = -\frac{3}{4}(x - 2)$ ✓

Midpoint $= (2, 8)$. Segment gradient $= \frac{4}{3}$, so the perpendicular gradient is $-\frac{3}{4}$. Equation:
 $y - 8 = -\frac{3}{4}(x - 2)$.

128 Find the equation of the perpendicular bisector of the segment joining $(5, 4)$ and $(-1, 12)$.

a. $y - 8 = \frac{3}{4}(x - 2)$ ✓

b. $y - 8 = -\frac{4}{3}(x - 2)$

c. $y - 8 = -\frac{3}{4}(x - 2)$

d. $y - 2 = \frac{3}{4}(x - 8)$

Midpoint $= (2, 8)$. Segment gradient $= -\frac{4}{3}$, so the perpendicular gradient is $\frac{3}{4}$. Equation:
 $y - 8 = \frac{3}{4}(x - 2)$.

129 Find the equation of the perpendicular bisector of the segment joining (4, 5) and (−2, 13).

a. $y - 1 = \frac{3}{4}(x - 9)$

b. $y - 9 = \frac{3}{4}(x - 1)$ ✓

c. $y - 9 = -\frac{3}{4}(x - 1)$

d. $y - 9 = -\frac{4}{3}(x - 1)$

Midpoint = (1, 9). Segment gradient = $-\frac{4}{3}$, so the perpendicular gradient is $\frac{3}{4}$. Equation:
 $y - 9 = \frac{3}{4}(x - 1)$.

130 Find the equation of the perpendicular bisector of the segment joining (2, 0) and (14, −16).

a. $y - 8 = \frac{3}{4}(x - (-8))$

b. $y - (-8) = -\frac{4}{3}(x - 8)$

c. $y - (-8) = -\frac{3}{4}(x - 8)$

d. $y - (-8) = \frac{3}{4}(x - 8)$ ✓

Midpoint = (8, −8). Segment gradient = $-\frac{4}{3}$, so the perpendicular gradient is $\frac{3}{4}$. Equation:
 $y - (-8) = \frac{3}{4}(x - 8)$.

131 Find the equation of the perpendicular bisector of the segment joining (2, −2) and (−10, −18).

a. $y - (-10) = -\frac{3}{4}(x - (-4))$ ✓

b. $y - (-4) = -\frac{3}{4}(x - (-10))$

c. $y - (-10) = \frac{3}{4}(x - (-4))$

d. $y - (-10) = \frac{4}{3}(x - (-4))$

Midpoint = (−4, −10). Segment gradient = $\frac{4}{3}$, so the perpendicular gradient is $-\frac{3}{4}$. Equation:
 $y - (-10) = -\frac{3}{4}(x - (-4))$.